higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

INTRODUCTION TO INFORMATION SYSTEMS
NQF LEVEL 2

1 MARCH 2019

This marking guideline consists of 12 pages.

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NOTE TO MARKERS

1. This memorandum should be used as a guide and is not exhaustive of possible solutions.

2. The candidate is not expected to answer verbatim as per memorandum in SECTION B.

3. Should a candidate's response be relevant to the question, but not the same as the memorandum, the mark may be awarded in line with the mark scheme provided.

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SECTION A

QUESTION 1.1

1.1	1.1.1	B
	1.1.2	A
	1.1.3	D
	1.1.4	A
	1.1.5	D
	1.1.6	C
	1.1.7	A
	1.1.8	D
	1.1.9	B
	1.1.10	C

(10 × 1) (10)

QUESTION 1.2

1.2	1.2.1	Organisation
	1.2.2	Knowledge
	1.2.3	Mandatory
	1.2.4	Virus
	1.2.5	Storage
	1.2.6	Application
	1.2.7	Terminal
	1.2.8	Output
	1.2.9	Optical
	1.2.10	Master

(10 × 1) (10)

1.3	1.3.1	True
	1.3.2	True
	1.3.3	True
	1.3.4	False
	1.3.5	True
	1.3.6	True
	1.3.7	False
	1.3.8	False
	1.3.9	False
	1.3.10	False

(10 × 1) (10)

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1.4	1.4.1	G		
	1.4.2	A		
	1.4.3	I		
	1.4.4	B		
	1.4.5	J		
	1.4.6	D		
	1.4.7	F		
	1.4.8	C		
	1.4.9	H		
	1.4.10	E		
			(10 × 1)	(10)

1.5	1.5.1	Chief executive officer		
	1.5.2	Personal Computer		
	1.5.3	Arithmetic-Logic Unit		
	1.5.4	Decision Support System		
	1.5.5	Advanced micro devices		
	1.5.6	Dynamic random access memory		
	1.5.7	Windows media audio		
	1.5.8	Computer-aided design		
	1.5.9	Electronic Funds Transfer		
	1.5.10	Digital versatile disk		
			(10 × 1)	(10)
				[50]

TOTAL SECTION A: 50

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SECTION B

QUESTION 2

2.1 2.1.1
- Operations manager – Hleza
- Because of the production statistics that he has to do, that makes him the manager for all the operations of the business.

(Any other relevant 2 × 1) (2)

2.1.2
- Financial manager – Dlamane
- He is the one that has to do the annual report on the income and expenditure for the year. (Any other relevant 2 × 1) (2)

2.1.3
- Research and development manager – Siphiwa
- He is the one that is responsible for the finding the right equipment that is currently being offered on the market and present his findings on his research. (Any other relevant 2 × 1) (2)

2.1.4
- Human Resources manager – Mpanza
- He is to overlooks all personnel matters relating to staffing and salaries. (Any other relevant 2 × 1) (2)

2.1.5
- Marketing manager – Sunitha
- She is to bring the representative from the advertising agency.

(Any other relevant 2 × 1) (2)

2.2 Is the process of planning, organising, staffing, leading and controlling to achieve a specific goal. (1 × 2) (2)

2.3 Methodical is being systematic✓ when analysing the information, having a step-by-step approach✓ regarding what to do and how to do it ***while*** analytical is the ability to look at all the information✓ and breaking it down into various components.✓ (4)

[16]

QUESTION 3

3.1 A –Information systems specialists develop and support an organisation's, ✓ computer technology and database systems ✓
B - An end user is the person ✓ that a software program or hardware device is designed for (They use the software) ✓
C – Processing device/s is any component in a computer ✓that manages the storage and retrieval of information ✓
D – System software is a type of computer program typically the OS ✓ that is designed to run and manage a computer's hardware and application programs. ✓
E – Network hardware are physical devices which are required for communication ✓and interaction between devices on a computer network ✓

(5 × 2) (10)

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3.2 A – Information systems specialists
E.g. Analysts, programmers, software testers

B - An end user
E.g. Secretary, Sales person, POS operator, Clerk etc. Home user

C – Processing device/s
E.g. Central processing unit (CPU), Graphics processing unit (GPU), Motherboard. Network card. Sound card. Video card

D – System software
E.g. Operating System (Also accept the name of an operating system such as Windows, Android), BIOS, Boot loader, assembler, device driver, utilities software (e.g. backup or systems restore)

E – Network hardware
E.g. gateways, routers, network bridges, modems, wireless access points, networking cables, line drivers, switches, hubs, and repeaters
(Any one acceptable example per component)(1 × 5) (5)
[15]

QUESTION 4

4.1 Centralised information system (1)

4.2
- Duplicate of information is eliminated
- The same information is shared among all users.
- There is improved security control as data sets are stored in one place.
- All related information is stored in one place.
- Responsibility and accountability are easily placed or assigned.

(Any 3 × 1) (3)

4.3
- Designing and setting up the system are complex.
- There is less flexibility in the system.
- When the network is down, communication among terminals is affected. (3)

4.4
- TotalAV
- AVG
- PcProtect
- Norton
- McAfee
- Avira
- Avast

(Any acceptable example). (1)

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4.5
- System or boot virus – attached themselves to a disk or hard disk, when the operating system is running, files on the external device, this can infect the system. Booting the system will than trigger the virus. ✓✓
- Macro virus – especial in Ms Word or Ms excel, they are designed to corrupt, erase or share data.
- Polymorphic virus – encrypt or encode themselves in different ways making it difficult even for anti-viruses to find them.
- Worm – not a virus but similar to a virus, slows things down as they consume more and more bandwidth, they maybe be spread by sending emails from one's account to the other recipient's computer.
- Ransomware – type of a malware that takes your file hostage by encrypting the file so that you are unable to access them.
- Trojan – work with computer viruses to exploits, steal and destroy sensitive information.

Allocate 1 mark for the type and one mark for the description. (1+1 × 3) (6)

4.6 The need to keep the anti-virus updated is the fact that there are new viruses that discovered, so it is important to keep it up to date as it will fight and disinfect the latest types of viruses. (Any other relevant 2 × 1) (2)

4.7
- Firewalls
- Passwords
- Biometrics
- Intrusion-detection system
- Emergency powered backup
- Encryption
- Backups (Any 5 × 1) (5)

[21]

QUESTION 5

5.1 General purpose computers are ordinary computers with ordinary software. ✓ They are capable of performing a wide range of tasks and can be used for word processing✓, web browsing and several other tasks while special-purpose computers are designed to perform special tasks only✓. They cannot be used for any other tasks and special software must be installed to them to function.✓ (Any other relevant answer) (4)

5.2 5.2.1 Takealot (1)

5.2.2 General purpose (1)

5.2.3
- Notebook is more compact than a Laptop
- Notebook is often lighter than a laptop
- Notebook is more portable
- Due to its size a notebook may not always have additional peripherals such as an optical drive (Any other relevant 2 × 1) (2)

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5.2.4 128 GB (1)

5.2.5 Solid State Drive (1)

5.2.6 13.3” Inches (1)

5.2.7
- No
- There is no optical drive included (2)

5.2.8 Using Wi-Fi (1)

5.2.9
- Screen (it is a touch screen)
- Web Camera
- Microphone (Any 1 × 2) (2)

5.2.10 Speakers (1)

5.2.11 The user may use (try) the product for 30 days ✓ and then needs to purchase a license for it to use it further. ✓ (2)

5.2.12 Intel Core i5-7Y54 (2)

5.4
- Fetch
- Decode
- Execute
- Write back / Store (4)

[25]

QUESTION 6

6.1
- A – Fingerprint reader/scanner
- B – Touch pad
- C – Web cam (3)

6.2
- Keyboard entry uses keystrokes to enter data into a system.
- Direct entry feeds data into the computer by reading data from magnetic media or paper such as a barcode or RFID tag, there are no keystrokes.

(2 × 2) (4)

6.3 QWERTY (1)

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6.4 Real time Processing ✓
Real time systems process data immediately and produce the output before the next input. ✓✓

Examples for real time processing
Booking of flight ticket
Patient monitoring in the ICU - Computer alerts the nurse immediately if any of the patient condition goes abnormal.
Real time processing example - Flight booking

1 Mark for an applicable example

Online / Interactive processing ✓
An online processing is where the user is directly interacting with a remote computer. User input data directly into the system and then it is processed. ✓✓

Example for online processing:
- Booking a flight ticket using internet.
- Booking a cinema ticket.
- Ordering for a book online.
- Managing a bank account online

1 Mark for an applicable example

Source (http://www.icteducatoronline.com /index.php?option=com_content&view=article&id=14&Itemid=147) (8)

6.5
- AGP provides a dedicated pathway between the slot and the processor rather than sharing the PCI bus.
- The direct connection allows higher clock speeds.
- AGP has improved overall data throughput. (Any 1 × 2) (2)

[18]

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QUESTION 7

7.1 7.1.1 HDD – Hard disk drive (1)

7.1.2 Magnetic (1)

7.1.3
- There's just a large shiny, circular "plate" of magnetic material called a platter,
- divided into billions of tiny areas.
- Each one of those areas can be independently magnetized (to store a 1) or demagnetized (to store a 0).
- Magnetism is used in computer storage because it goes on storing information even when the power is switched off. If you magnetize a nail, it stays magnetized until you demagnetize it

Source: https://www.explainthatstuff.com/harddrive.html

Or

- Hard disk stores information in the form of magnetic fields.
- Data is stored digitally in the form of tiny magnetized regions on the platter where each region represents a bit.
- To write a data on the hard disk, a magnetic field is placed on the tiny field in one of these two polarities: N-S – If North Pole arrives before the south pole and S-N – if the south pole arrives before the north pole while the field is accessed.
- An orientation in the one direction (like N-S) can represent the '1' while the opposite orientation (S-N) represents "0". This polarity is sensed by integrated controllers built within the hard disk

Source https://www.engineersgarage.com/mygarage/how-data-is-stored-on-hard-disk (4)

7.1.4
- Relies on moving parts
- Disk surface can be damaged
- Heavy power consumption
- Noisy
- Slower read and write speed than RAM
- Far slower to access data than ROM or RAM chips.
- Hard disks can crash.
- Regular crashes can damage the surface of the disk, leading to loss of data in that sector.
- The disk is fixed inside the computer and cannot easily be transferred to another computer. (Any acceptable 3 × 1) (3)

7.1.5 Platter (1)

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7.2
- System crashes or failures
- Natural disasters
- Loss of information
- Process failures
- Problems with a specific application
- Files deleted by mistake
- Theft of equipment (Any 5 × 1) (5)

7.3
- Photo copier
- Printer
- Scanner
- Fax / Scan to e-email (4 × 1) (4)

7.4 A laser printer would be suitable for the business because:
- It offers quality printing.
- Some laser printers can print in colour
- It is fast
- It has lower costs associated to it
- It is durable
- Has a small machine footprint (Any 4 × 1) (4)

7.5 1024 ✓ terabytes (TB) ✓

If the student stated 1000 TB only allocate 1 mark (2)
[25]

QUESTION 8

8.1 8.1.1 The statement is TRUE ✓

A basic computer performs Input processing output and storage operations. ✓✓
The smartwatch has input components, output components, processing components and storage components. ✓✓ (1 + 4) (5)

8.1.2 Liquid crystal display (1)

8.1.3 In an LCD display, the pixels are switched on or off electronically ✓ using liquid crystals to rotate polarized light ✓ to create colours. ✓ (3)

8.1.4 Plasma or LED / Light emitting diode - Technology screens (1)

8.1.5
- Music Player
- Calendar
- Email (Any 1 × 1) (1)

8.1.6 To store additional files and data ✓such as, picture files from the camera, applications, recordings and other files transferred ✓ (2)

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8.1.7 A soft copy is a document saved on a computer. It is the electronic version of a document, which can be opened and edited using a software program and is visible on a screen. (1 × 2) (2)

8.1.8 a SoftCopy Device (1)

8.2

8.2.1	B✓	III✓
8.2.2	A✓	V✓
8.2.3	D✓	I✓
8.2.4	C✓	II✓
8.2.5	E✓	IV ✓

(5 × 2) (10)

8.3 ***Impact printers*** make contact with the paper and usually from the printed image by pressing the inked ribbon against the paper using a hammer or pins **while *non-impact printers*** from characters and images without a direct physical contact between the paper and the printing mechanism. (1 × 2) (2)

8.11
- Inkjet printer
- Thermal printer
- Plotter

(Any 2 × 1) (2)

[30]

TOTAL SECTION B: 150
GRAND TOTAL: 200

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